MILITARY SPECIFICATION

COMPUTER, BALLISTICS: XM19

1. SCOPE

1.1 This specification covers one type of instrument known as Ballistics Computer: XM19, used to receive the log of range data from a rangefinder calibrated for range measurement in meters, and compute superelevation for various ammunitions and transmit same to a gun sighting system in the 90 MM Gun Tank: M48A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Material, Packaging of
MIL-F-14619	Fire Control Parts Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification for

FSC 1220

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8205605	Computer, Ballistics: XM19
IEL-8205605	Inspection Equipment List for Computer, Ballistics: XM19

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one ballistics computer, representative of production shall be made available to the Government to determine if the ballistics computer meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The computer shall be manufactured in accordance with Drawing F8205605, "Ballistics, Computer: XM19" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

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3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

3.5 Performance.

3.5.1 Output calibration.

3.5.1.1 Zero reference.- Input shaft rotation shall be measured from a precise 1200 meter (zero reference) check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

3.5.1.2 1200 meter check point.- With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for ballistics drive, the ammunition selector shall be set on the HEAT-T300E56 Cam. With the range correction knob set to zero and with the input shaft set at the 1200 meter check point, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width by approaching this range first from one direction then from the opposite direction, without over travel. Note the position of the output shaft for ballistics drive and split the backlash error at this output shaft. This is then the check point for computer output. The position of the superelevation counter shall be established from this check point and shall conform to the requirements of Table I.

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TABLE I

Position of Superelevation Counter				
Output shaft position	277.92° CCW	1200 meter check point 0°	3R-312.48° CW	12R-322.48° CW
Tolerance	± 8°	± 3°	± 8°	± 8°
Superelevation counter mils	1.80	5.66	25.00	70.00

3.5.1.3 Outer index and inner index.- Using the HEAT-T300E56 Cam only, position the input shaft to correspond with each of the ranges as indicated in Table II and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of this output shaft shall correspond to the value in Tables II and III. This procedure shall be repeated to check each cam at 1200 meters only. All checks shall be made by approaching index from clockwise and counterclockwise direction.

3.5.1.4 Backlash.- Backlash of the output shaft for the superelevator shall not exceed a total of one degree at any point of rotation of the shaft.

3.5.1.5 Outer index and range scale.- Using the HEAT-T300E56 Cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in Table II, at which point the rotation of the output shaft for ballistics drive shall correspond to the values in Tables II and III. All checks shall be made by approaching index from clockwise and counterclockwise direction.

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3.5.1.6 Servo follow-up sensitivity.- With an output shaft load of between 75 and 90 ounce inches applied to the output shaft for the ballistics drive, the ammunition selector set for the HEAT-T300E56 Cam, the power on at 18 plus or minus 0.5 volts and the automatic follow-up cycle completed at ranges between 3600 and 4400 meters, rotate the input shaft slowly so that it is possible to count the number of times the contact closes in order to advance the output shaft to the ballistics drive by approximately 72 degrees. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposute direction. The average allowable superelevation increment for each range shall not exceed 10 degrees 45 minutes and the maximum superelevation increment shall not exceed 21 degrees 30 minutes as read at the output shaft to the ballistics drive.

3.5.2 Power operation.- With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for the ballistics drive, the rotation of this output shaft shall correspond to the values indicated in Tables II and III for the ammunition and ranges specified therein. This requirement shall be met by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 plus or minus 0.5 volts. The computer shall also meet the accuracy requirement of Tables II and III for the following ammunition and at ranges indicated when checked at 18 plus or minus 0.5 volts.

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Ammunition and Range for 18 Volt Check

Ammunition	Range
HEAT-T300E56	500, 1000, 2000 and 3600 meters

All checks shall be made approaching index from both clockwise and counterclockwise direction.

TABLE II

Computer accuracy as determined by postion of rotation of the Output Shaft based on 1200 meter reference range for HEAT-T-300E56 Cam (28 Plus or Minus 0.5 volts)

Range meters	500	1000	1200	1500	2000	3000	3600	4300	4400
Input shaft position position tolerance ± 1°	8R 330.48° CW	1R 308.52° CW	0° Check Point	2R 98.25° CCW	5R 73.08° CCW	9R 119.88° CCW	11R 68.40° CCW	13R 0° CCW	13R 84.24° CCW
HEAT-T300E56	277.92° CCW *1.80	88.56° CCW *4.43	0° *5.66	150.38° CW *7.75	1R 110.80° CW *12.20	4R 236.16° CW *28.94	8R 236.16° CW *48.94	17R 287.28° CW *88.99	
HE-T, M71 / WP-M313	131.04° CCW *3.84	160.56° CW *7.89	285.96 CW *9.59	1R 112.32° CW *12.22	2R 85.68° CW *16.85	4R 110.16° CW *27.19	5R 254.16° CW *34.19		17R 263.68° CW *44.60
AP-T, M318A1	259.92° CCW *3.05	51.84° CW *6.38	152.64° CW *7.78	310.32° CW *9.97	1R 230.40° CW *13.86	3R 128.16° CW *22.64	4R 220.32° CW *28.76		6R 173.16° CW *38.11

* = Corresponding nominal superelevation Mil value (see 6.6)

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TABLE III

Output Shaft Tolerance		
A Output shaft rotation	B Degrees	C Mils*
0 up to but less than 6	± 7.20	± 0.10
6 up to but less than 11	± 10.80	± 0.15
11 up to but less than 26	± 14.40	± 0.20
26 up to but less than 31	± 18.00	± 0.25
31 up to but less than 36	± 25.20	± 0.35
36 up to but less than 41	± 28.80	± 0.40
41 up to but less than 61	± 36.00	± 0.50
61 up to but less than 86	± 43.20	± 0.60
86 up to but less than 98.5	± 50.40	± 0.70

* The data contained in column "C" is informational only. (see 6.6)

3.5.3 Output shaft tolerance.- When the output shaft rotation of the computer, as read in degrees is equivalent to the superelevation value in mils as shown in column "A", the output shaft tolerance shall be within the corresponding value as shown in column "B".

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3.5.4 Stability.- Using 28 plus or minus 0.5 volts with a load of between 15 and 18 ounce inches applied to the output shaft to ballistics drive, ballistics correction set to introduce minus 15 percent within 1 percent, the rotation of the input shaft between minimum range and 1200 meters shall not cause the computer servo to display any self-sustaining or oscillatory operation. This check is required for all cams. A maximum of eight changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These eight changes may be counted as clicks.

3.5.5 Component operation.

3.5.5.1 Operational values.- Operational values shall not exceed those given in Table IV at the temperatures indicated.

TABLE IV

Operational Values			
Control	Operation	Standard ambient temperature to plus 150°F.	Minus 40°F.
Input Shaft	Rotate	16 oz. inches	45 oz. inches
Range Correction Knob	Pull Out	20 lbs.	24 lbs.
	Rotate	34 oz. inches	45 oz. inches
ammunition Selector	Rotate	60 lb inches	80 lbs inches
	Translate	10 lbs.	20 lbs.
Handcrank	Hold Engaged	12 lbs.	20 lbs.
	Rotate Engaged	6 lb. inches	20 lbs Inches
Reset Button	See Par. 3.6.3	15 lbs.	40 lbs.
Iris Dimmer	Rotate	Mechanically Operable	
Interrupter Switches	See Par. 3.6.3	Mechanically and Electrically Operable	

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3.5.5.2 Time of response.- At standard ambient temperature (plus 60 to 90 degrees Fahrenheit) with the output shaft load as specified in Paragraph 3.5.2 applied, the inner and outer indexes positioned to the values of Table V, and the ammunition selector set on the HEAT-T300E56 Cam, the outer index shall stabilize on the inner index within the elapsed time specified after power has been applied.

TABLE V

Time of Response			
Outer Index Position (Meters)	Inner Index Position (Meters)	Voltage (Plus or minus 0.5V)	Total Elapsed Time (Seconds)
4400	500	28	4
500	4400	28	4
500	4400	18	8
4400	500	18	8

3.5.5.3 Circuit breaker.

3.5.5.3.1 Breaker opens.- With the motor at standard temperature with 28 plus or minus 0.5 volts applied, the two indexes separated by at least 200 meters, the output shaft for the ballistics drive locked(immovable), the circuit breaker shall open in 90 seconds or less.

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3.5.5.3.2 Breaker closed.- At standard ambient temperature with the load as specified in Paragraph 3.5.2 applied to the output shaft for the ballistics drive, the computer operating at 28 plus or minus 0.5 volts, using the HEAT-T300E56 Cam, the input shaft shall be oscillated for a period of forty-five seconds over a spread of 1000 meters, plus or minus 100 meters between ranges of 1200 to 2800 meters, at a rate of 4 to 4.5 revolutions per second, during which time the circuit breaker shall not open.

3.5.5.4 Hand crank and input shaft travel.

3.5.5.4.1 The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 mils plus or minus 0.1 mil to plus 98.5 mils plus or minus 0.5 mil as indicated on the superelevation counter when the ballistics correction knob is set at zero.

3.5.5.4.2 The range input shaft shall rotate through an excursion of 24.0 revolutions plus 0.1 revolution and shall cover the range from 480 to 4575 meters within this excursion, as measured from the 1200 meter check point of the input shaft (see 6.7).

3.5.6 Superelevation cams.- The HEAT-T300E56 cam shall be placed in the computer followed by the HE-T-M71 (WP, M313) cam and the AP-T, M318A1 cam, followed by three spaces (see 6.5). The computer shall be shipped with the same cams with which it qualified.

3.6 Electrical check.

3.6.1 Insulation breakdown.- Insulation breakdown shall be in accordance with Specification MIL-F-13926 except that breakdown voltage of 500 volts RMS shall be substituted for 750 volts RMS.

3.6.2 Insulation resistance.- Insulation resistance shall be in accordance with Specification MIL-F-13926 except that the resistance value shall not be less than 2 megohms and a dc potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

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3.6.3 Continuity.- The electrical circuits shall be checked for continuity, and shall conform to the wiring diagram. When the hand crank is engaged the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammunition selector interrupter switch to open the motor circuit. Resetting of the handle shall cause switch to complete the circuit.

3.6.4 Illumination.- The internal lighting device shall provide adequate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened inclosure.

3.7 Environmental.

3.7.1 Storage temperatures.- The computer shall show no evidence of physical failure and a recheck of the computer shall meet the performance requirements of Paragraph 3.7.2.1 at standard ambient temperature after having been exposed and thermally stabilized at ambient temperatures of plus 160 and minus 80 degrees Fahrenheit.

3.7.2 Operating temperature.- When exposed and thermally stabilized at ambient temperatures of plus 150 and minus 40 degrees Fahrenheit the computer shall meet the following requirements:

(a) Applicable requirements of 3.5.4.1 at temperatures indicated in Table IV.

(b) At minus 40 degrees Fahrenheit the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 plus or minus 0.5 volts.

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(c) At plus 150 degrees Fahrenheit using 28 volts, plus or minus 0.5 volts, the ammunition selector set on the HEAT-T300E56 Cam, with a load of between 14 and 18 ounce inches applied to the output shaft for the ballistics drive, and with the indexes separated by 3600 meters, the outer index shall stabilize on the inner index and shall meet the applicable requirements of 3.5.3 when the switch is thrown to the "ON" position. Immediately upon completion of the environmental requirements of 3.7, the interior of the computer shall be dried by clean, warm air of between approximately plus 100 degrees to 140 degrees Fahrenheit.

3.7.2.1 Post environmental check.- Each computer which has been subjected to the environmental conditions of 3.7.1 and 3.7.2 shall meet the following requirements after thermal stabilization has been re-established at standard ambient temperature.

(a) The requirements of Table I and 3.5.4

(b) When checked under power operation at 28 volts, plus or minus 0.5 volt, with 75 to 90 ounch inch load applied, the output shaft for the ballistics drive shall indicate the rotational values of the Table II and within the tolerances of Table III for the following ammunition and ranges:

AMMUNITION	RANGES
HEAT-T300E56	500, 1000, 2000 and 3600 meters

3.7.3 Vibration.- With the ammunition selector set on the HEAT-T300E56 Cam, the outer index set at 3000 meters, the computer (less any shock mounts and brackets, Universal Assembly B8200862 and Output Shaft B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.7.3.1 Post vibration.- Subsequent to vibration the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall meet the requirements of this specification.

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3.7.4 Sealing.- The computer shall meet the requirements of the rain test of Specification MIL-F-13926.

3.8 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General requirements.- Inspection shall be in accordance with General Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one computer shall be selected by the Government inspector as soon as practicable after the start of production and shall be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- Contractor's inspection and contractor's inspection system shall be in accordance with Specification MIL-I-45208.

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4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table VI.

TABLE VI - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Inspection equipment shall conform to drawings listed in IEL-F8205605, except as provided for under the alternative inspection procedures and inspection equipment requirement of Specification MIL-I-45208. Provisions of Specification MIL-I-45208 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

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4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance,except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables VII and VIII shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE VII - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Zero reference	3.5.1.1	4.8.1	Visual
102. 1200 meter check point	3.5.1.2	4.8.3	Device, Special Testing (DST)
103. Outer index and index	3.5.1.3	4.8.4	DST
104. Backlash	3.5.1.4	4.8.5	DST
105. Outer index and range scale	3.5.1.5	4.8.6	DST
106. Servo follow-up sensitivity	3.5.1.6	4.8.7	DST
107. Power operation	3.5.2	4.8.8	DST
108. Stability	3.5.4	4.8.9	DST
109. Time of response	3.5.5.2	4.8.11	DST
110. Circuit breaker opens	3.5.5.3.1	4.8.12.1	DST
111. Circuit breaker closed	3.5.5.3.2	4.8.12.2	DST
112. Hand crank travel	3.5.5.4.1	4.8.13.1	DST
113. Input shaft travel	3.5.5.4.2	4.8.13.2	DST
114. Superelevation cams	3.5.6	4.8.14	DST
115. Insulation breakdown	3.6.1	4.8.15.1	Standard Measuring Equipment (SME)
116. Insulation resistance	3.6.2	4.8.15.2	SME
117. Continuity	3.6.3	4.8.15.3	SME
118. Illumination	3.6.4	4.8.16	Visual
119. Physical failure, post vibration	3.7.3.1	4.8.19.1	Visual-Tactile
120. Identification marking	3.8	4.9	Visual
MINOR: NONE DEFINED			

NOTE: The above tests shall be performed at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

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TABLE VIII - CLASSIFICATION OF DEFECTS

Use Inspection Level L-7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Operating temperature	3.7.2	4.8.17.2	Device, Special Testing (DST)
102. Outer index and inner index	3.7.2(b)	4.8.17.2	DST
103. Component operation	3.7.2(c)	4.8.17.2	DST
104. Superelevation counter, post environmental	3.7.2.1	4.8.18	DST
105. Stability, post environmental	3.7.2.1	4.8.18	DST
106. Power operation, post environmental	3.7.2.1	4.8.18	DST
MINOR: NONE DEFINED			

NOTE: The above "post environmental" tests shall be performed at standard ambient temperature, plus 60 degrees to plus 90 degrees Fahrenheit, except Tests 101, 102 and 103 which shall be performed at minus 40 degrees and plus 150 degrees, Fahrenheit.

4.8. Test methods and procedures.

4.8.1 Zero reference.- The 1200 meter check point as determined previously by the manufacturer shall be visually checked. It shall be indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal plate cover.

4.8.2 The following tests shall be conducted using Test Fixture A8565298 and in accordance with the operating instructions specified on Drawing F8565296 using appropriate metric scales. The computer input shaft and the fixture input mechanism shall be oriented to the 1200 meter check point.

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4.8.3 1200 meter check point.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the mil value specified in Table I is indicated on the computer superelevation mil counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in Table I.

4.8.4 Outer index and inner index.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in Table II, and the computer outer and inner indexes are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerances as specified in Table III. This test shall be repeated to check each cam at 1200 meters only.

4.8.5 Backlash.- Backlash of the computer superelevator output shaft shall not exceed the tolerance when measured at four (4) points of shaft rotation. The four (4) points of measure shall be approximately ninety degrees (90°) apart.

4.8.6 Outer index and range scale.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the computer outer index is coincident to the selector range mark on the computer range scale, the rotational position of the output shaft for the ballistics drive shall correspond to the values of Table II and not exceed the tolerances of Table III.

4.8.7 Servo follow-up sensitivity.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index to a range between 3600 meters and 4400 meters. With the computer in power operation, the average and maximum recorded increment change (closing of computer servo contacts) resulting from a total rotation of the output shaft 72° clockwise and 72° counterclockwise shall not exceed the tolerance specified. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.8.8 Power operation.- This test shall be conducted as specified on Drawing F8565296 with the range correction knob set to zero.

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4.8.8.1 With the input voltage set at 28 plus or minus 0.5 volts and the circuit breaker switch in the "ON" position, the accuracy of each cam shall be tested at the ranges listed in Tables II for computer output shaft rotation. When the input shaft is rotated to a value indicated in Table II for a given range, the rotational position of the output shaft shall correspond to the value for that range in Table II within the tolerances specified in Table III.

4.8.8.2 With the input voltage set at 18 plus or minus 0.5 volts, the computer shall meet the accuracy requirements of Tables II and III for the cams and ranges specified.

4.8.9 Stability.- Prepare the computer for this test in accordance with Drawing F8565296; set the range correction knob to minus fifteen percent (-15%) within one percent (1%), and adjust the input voltage to 28 plus or minus 0.5 volts. Rotate the computer inner index 1200 meters; throw the circuit breaker switch to "ON" position to energize the servo motor. The outer index shall "settle out" (cease oscillatory operations) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo. The inner index shall then be rotated to four (4) different ranges between 1200 meters and minimum range, and the outer index again shall "settle out" coincident to the inner index within eight (8) changes in direction of the servo. Set the range correction knob to plus fifteen percent (+15%) within one percent (1%), and repeat the aforementioned test. All cams installed in or accompanying the computer shall be subjected to this test.

4.8.10 Component operation tests.- Operating tests shall not exceed the values listed in Table IV at the temperatures indicated.

4.8.11 Time of response.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized, the computer outer index shall stabilize on the inner index within the time cycle specified in Table V.

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4.8.12 Circuit breakers.

4.8.12.1 Circuit breaker, opens.- The inner and outer index shall be separated by the specified distance, and the output shaft for the ballistics drive immobilized. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized, the circuit breaker shall open (switch move to "OFF" position) within the specified time cycle.

4.8.12.2 Circuit breaker, closed.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index coincident on 2000 meters of range. The computer input shaft shall be oscillated as specified, with the circuit breaker switch in the "ON" position and the servo motor energized. The circuit breaker shall not open (switch move to "OFF" position) as a result of this oscillation.

4.8.13 Hand crank and input shaft travel.

4.8.13.1 The test for hand crank travel shall be conducted in accordance with Drawing F8565296, and shall meet the specified requirements.

4.8.13.2 The test for input shaft travel shall be conducted in accordance with Drawing F8565296, and shall meet the specified requirements.

4.8.14 Superelevation cams.- The inspection sequence and arrangement of cams in the computer shall be as specified in Drawing F8565296.

4.8.15 Electrical check.- The following test shall be conducted at any time subsequent to vibration. Before testing, remove lamp bulbs and disconnect capacitor from computer circuit.

4.8.15.1 Insulation breakdown.- This test shall be conducted in accordance with Specification MIL-F-13926 and shall meet the specified requirements. Procedure I shall apply.

4.8.15.2 Insulation resistance.- This test shall be conducted in accordance with Specification MIL-F-13926 and shall meet the specified requirements.

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4.8.15.3 Continuity.- A continuity test of the computer circuits shall be made using standard measuring equipment to determine conformance with the wiring diagram. Actuation of the hand crank, reset button, and ammunition selector handle shall cause the computer to function.

4.8.16 Illumination.- The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements when observed in a darkened enclosure with the unaided eye.

4.8.17 Environmental.

4.8.17.1 Storage temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 to the conditions specified. Upon return to standard ambient temperature, the computer unit shall be visually inspected for damage and then subjected to the tests specified in 4.8.18. This test shall be conducted on the production model only.

4.8.17.2 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 to the conditions specified. The following tests shall be performed at the temperatures specified.

(a) The controls listed in Table IV shall not exceed the applicable torque and force values at the temperatures specified therein for the operation tests of those controls. The equipment used for these tests is listed on Drawing D8570433. The reset button shall be inspected using standard measuring equipment capable of measuring within one-quarter (1/4) pound.

(b) Rotate the input shaft from the clockwise stop to the counterclockwise stop; observe the outer index; it shall follow the inner index in the required time limit at the specified voltage. Repeat the test by rotating the input shaft in the opposite direction. The circuit breaker switch shall be in the "ON" position. This test shall be performed at minus 40 degrees Fahrenheit.

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(c) Set the ammunition selector on the cam specified, the range correction knob to zero, and apply the specified torque load to the computer output shaft. Position the computer inner index at 500 meters and the outer index at 4400 meters on the range scale. When the circuit breaker switch is thrown to the "ON" position, the outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 meters and the inner index positioned at 4400 meters on the range scale.

4.8.18 Post environmental check.- Following the environmental test, the computer shall be thermally stabilized at room temperature and tested for the following:

(a) The tests of 4.8.3 and 4.8.9.

(b) With the required load applied to the output shaft and at the specified voltage, the computer shall be tested in accordance with 4.8.8 and 4.8.8.1.

4.8.19 Vibration.- Each computer in the sample shall be mounted on Fixture F7650085 using Adapter F7659608 and vibrated in accordance with the requirements specified.

4.8.19.1 Post vibration, physical failure.- Following vibration, the computer shall be subjected to a visual and tactile examination for damage such as loose, bent or broken parts, etc. The computer shall then be subjected to all tests of this specification.

4.8.20 Sealing (water resistance).- This test shall be conducted in the accordance with Procedure I of Specification MIL-F-13926. The computer shall show no evidence of water penetration as a result of this test. The receptacle shall be covered with waterproof tape during this test. The computers that are opened for environmental drying and examination for water penetration shall be resealed to meet the requirements of the drawings prior to submission for final acceptance.

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4.9 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.10 Packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

4.11 Final acceptance.- Final acceptance of the lot of computers shall be withheld by the Government inspector until the sample lots of computers, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The computer shall be packaged, packed and marked in accordance with Specification MIL-F-14619-P8205605. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Ballistics Computer: XM19 is an electromechanical component of the primary fire control system for 90 MM Gun Tank: M48A3. The ballistics computer manually or electrically determines and relays to the sighting system the amount of gun superelevation required for specific target range, and ammunition type.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1).

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6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request or the contractor.

6.4 Definitions.

6.4.1 Special testing device(DST).- Special testing devices are defined as special inspection fixtures, instruments and modified standard measuring equipment.

6.4.2 Standard measuring equipment(SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.5 Cam designation.- The designation HE-T, M71 and designation WP, M313 both appear on the same cam, the ammo tab for this cam will have both designations.

6.6 Superelevation values in mils.- The mil values in Table II represent the corresponding nominal superelevation values in mils for the calculated rotational values in the table. The actual superelevation value in mils that will correspond to computer output rotation will be obtained by modifying the nominal value by the relationship that each 7.20 degrees of rotation of the output shaft above or below the nominal value is equal to 0.10 mil.

6.7 Input shaft excursion

From 1200 meters 0° check point
9R 119.97° CW= 480 meters
13R 84.38° CCW = 4400 meters
13R 227.40° CCW = 4575 meters

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